

- Q.25 What are the methods used for inductance measurements? Explain any one of them.
- Q.26 Explain the construction details of energy meter.
- Q.27 Describe phase sequence indicators.
- Q.28 Explain the working of Maxwell bridge with the help of diagram.
- Q.29 Explain how frequency is measured using CRO?
- Q.30 Explain two wattmeter power measurement method in details.
- Q.31 Draw and explain CRT.
- Q.32 Explain Q meter and also write at least two applications of it.
- Q.33 How does a DC voltage measure using a multimeter?
- Q.34 What are the classification of instruments?
- Q.35 Write at least five applications of CRO.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw and explain the block diagram of DSO. Write its applications.
- Q.37 Explain the block diagram showing the main components in a digital storage oscilloscope what are the main differences between basic CRO and dual slope oscillator?
- Q.38 Explain the working principle of rectifier type meter with suitable diagram. What is the main difference between moving iron and PMMC instrument?

(100) (4) 181533/121533/031533

No. of Printed Pages : 4 181533/121533/031533
Roll No.

3rd Sem / Instrumentation & Control Sub : Test and Measuring Instruments

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Meter is used for the measurement of _____.
a) Inductance
b) Capacitance
c) Resistance
d) Characteristics of capacitors
- Q.2 The moving coil in electro-dynamometer is also called as _____.
a) Fixed Coil b) Pressure Coil
c) Current d) None of these
- Q.3 Henry is the unit of _____.
a) Inductance b) Resistance
c) Energy d) None of the above
- Q.4 Which of the following cannot be used for measurement of resistance?
a) Wheatstone bridge b) Potentiometer
c) Megger d) None of the above

(1) 181533/121533/031533

- Q.5 CRO stands for
- Cathode right oscilloscope
 - Cathode ray oscilloscope
 - Cathode ray oscillations
 - None of these
- Q.6 Unit of capacitance is
- Henry
 - Weber
 - Fared
 - Joule
- Q.7 Phase sequence indicator is used for _____ measurements.
- Phase
 - Frequency
 - Capacitor
 - Energy
- Q.8 An LCR bridge meter is a specialized electronic test instrument designed to measure the inductance, capacitance and resistance components.
- Q meter
 - Wheatstone bridge
 - Hay's bridge
 - LCR bridge
- Q.9 PMMC stands for
- Permanent magnet merging coil
 - Permanent magnet moving coil
 - Phase magnet moving coil
 - Phase magnet merging coi
- Q.10 Which of the following is used for the measurement of frequency
- Stroboscope
 - Maxwell bridge
 - LCR bridge
 - None of the above

(2) 181533/121533/031533

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Expand CRO
- Q.12 A Moving iron meter measures only AC. (True / False)
- Q.13 Ohmmeter is used only for the measurement of inductance. (True / False)
- Q.14 Wagner earth device is a circuit used for the elimination of stray capacitance. (True / False)
- Q.15 Wattmeter Power measurement method uses only two watt meter. (True / False)
- Q.16 The output of the sweep generator is applied to the _____ plate of CRO.
- Q.17 Define Q meter.
- Q.18 Write one application of typical instrumentation.
- Q.19 Define measurement.
- Q.20 Write one application of energy meter.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain electrostatic type instrument with diagram.
- Q.22 Draw and explain the working of potentiometer for resistance measurement.
- Q.23 Write a short note on the need and importance of instruments.
- Q.24 Explain the working of stroboscopes.

(3) 181533/121533/031533